POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Advanced Programming with Java

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is the role of the JVM in Java Architecture? Describe the steps to compile and run a Java program. 7

   b) Write a multithreaded program in Java to print even and odd numbers alternatively. 8

OR

What are design patterns? Compare the Singleton pattern with the Factory pattern.

2. a) Define Interface. As we know, there is no room for multiple inheritance using classes. How do you achieve multiple inheritance in java. Explain with Example. 8

   b) What is a constructor in Java? Explain the different types of constructors with examples. 7

3. a) Explain the event handling mechanism in Java. Write a program that uses ActionListener to handle button clicks. 7

OR

Define stage and scene in JavaFX. Differentiate between Swing and JavaFx.

   b) Compare the different layout managers in Java Swing with examples for each. 8

4. a) Develop RMI application where client input two numbers if both numbers are even then result $(a+b)^2$ otherwise $(a+b)^3$. 8

   b) Write a Java program to implement a client-server application using TCP sockets. Explain the flow of communication. 7

5. a) What are the main components of JDBC architecture? Explain the role of the DriverManager and Connection interface. 7

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b) Write a Java program to establish a connection to a database using JDBC. Demonstrate the use of *Statement* and *PreparedStatement* objects with examples. 8

6. a) What are the differences between static and dynamic web applications? How does a servlet fit into a dynamic web application? 7

b) Discuss the life cycle of a servlet in detail, including the methods init(), service(), and destroy(). 8

7. Write short notes on: **(Any two)** 2×5

a) Handling GET/POST Requests

b) Hibernate

c) Advantages of MVC

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